

IBM University Engagement Project Submission

“Exploring the Power of Agentic AI with IBM Cloud and Orchestrate”

Project Title – Fitness Buddy Agent – Your AI-Powered Health & Fitness Companion

Domain of Project – Health & Fitness / Wellness

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Problem Statement -

Problem Statement : Fitness Buddy Agent

The Challenge

In today's fast-paced world, many individuals struggle to maintain a healthy lifestyle due to **lack of personalized guidance, time constraints, and inconsistent motivation**. Traditional fitness solutions often require expensive subscriptions, in-person consultations, or rigid schedules that don't adapt to personal preferences or daily routines. Without **an accessible, friendly, and intelligent virtual fitness assistant**, many struggle to stay consistent, motivated, and on track with their fitness goals.

The Objective

Build an **AI-powered Fitness Buddy Agent** that delivers **personalized fitness advice, healthy lifestyle suggestions, and basic nutrition guidance** for individuals, available on-demand and at any time. The solution should:

- **Home Workout Recommendations** – Recommend home workouts and routines based on user input, fitness level, and available time.
- **Motivational Coaching** – Provide motivational tips and daily fitness inspiration to keep users engaged and consistent.
- **Nutrition Guidance** – Suggest simple, nutritious meal ideas that complement users' fitness goals.
- **Habit Building** – Encourage consistency and long-term healthy habits through reminders and progress check-ins.
- **Multi-Agent System** –
 - **Workout Recommendation Agent** (suggests home workouts based on user input)
 - **Motivation & Inspiration Agent** (delivers daily motivational tips and fitness inspiration)
 - **Nutrition Guidance Agent** (suggests simple, nutritious meal ideas)
 - **Habit & Consistency Agent** (encourages habit-building and consistency)
- **Progress Dashboard** – Show daily/weekly workout activity, streaks, and progress toward fitness goals.

Proposed Solution -

Proposed Solution: AI-Powered Fitness Buddy Agent

The proposed solution is a **multi-agent AI-powered Fitness Buddy Agent** built using **IBM watsonx Orchestrate**, **IBM watsonx.ai**, and **IBM Granite Models** to provide personalized, on-demand fitness and lifestyle guidance. The system addresses challenges of inconsistent motivation, lack of personalized guidance, and rigid traditional fitness solutions through an intelligent, conversational architecture built entirely on IBM Cloud Lite services.

At its core, the solution uses a **Retrieval-Augmented Generation (RAG)** pipeline to fetch reliable fitness and nutrition knowledge (workout techniques, healthy meal ideas, motivational content) from trusted sources. This data is embedded and stored in a vector database, enabling the system to retrieve accurate, up-to-date guidance using Granite models.

The system consists of four key agents:

Workout Recommendation Agent recommends home workouts and routines tailored to the user's fitness level, goals, and available time.

Motivation & Inspiration Agent delivers daily motivational tips and fitness inspiration to keep users engaged and consistent.

Nutrition Guidance Agent suggests simple, nutritious meal ideas that complement the user's fitness routine.

Habit & Consistency Agent encourages habit-building through reminders, streak tracking, and consistency check-ins.

The solution includes a **user-friendly dashboard** (built with Streamlit or React) that visualizes daily and weekly workout activity, streaks, and progress toward fitness goals.

IBM watsonx Orchestrate is used to orchestrate agent workflows, while watsonx.ai ensures scalable model deployment, embeddings, and AI governance. Granite models power reasoning, summarization, and personalization.

Overall, this solution delivers an **end-to-end intelligent fitness and lifestyle assistant** that promotes healthier, more active lifestyles through personalized workouts, motivational support, and simple nutrition guidance.

Technology Used

- **IBM Granite model** - for eg – ibm-granite-3-2-8b – For natural language understanding, reasoning, and personalized fitness/nutrition guidance
- **IBM Cloud Lite** – Provides the free-tier infrastructure to build, deploy, and scale the Fitness Buddy Agent efficiently. IBM Cloud Lite enables a **reliable, secure, and scalable environment** for deploying the Fitness Buddy Agent at no cost, ideal for a student project.
- **RAG**– RAG to **fetch reliable workout and nutrition knowledge first, then generate intelligent, tailored answers**, making the Fitness Buddy Agent more trustworthy and effective.
- **IBM Watsonx.ai** - For model deployment, embeddings, and AI governance
- **Vector Database** (FAISS/Chroma) – For RAG-based retrieval of workout and nutrition knowledge

Orchestrate component Used -

- 1) **Chat Input** – Interface where users enter their fitness queries, preferences, or goals (e.g., "suggest a 20-min home workout" or "give me a healthy breakfast idea").
- 2) **IBM watsonx AI Agent** - Processes user input using Granite models to generate personalized workout, motivation, and nutrition recommendations.
- 3) **Name of IBM Granite model** - Granite-4.0-8B-Instruct → Fast, efficient for real-time fitness and nutrition recommendations
- 4) **Chat output** -Displays AI-generated responses such as workout routines, motivational tips, and healthy meal ideas.
- 5) **File component** - Stores and retrieves user preferences, workout history, and reference fitness/nutrition datasets for analysis and tracking.
- 6) **Knowledge Base / Search component** – Grounds responses in reliable fitness and nutrition reference content.

Agent Instructions – Fitness Buddy

Agent (IBM watsonx ai)

Building an AI-powered...

Edit agent | IBM watso...

AI Fitness Buddy Agen...

SnehaSiddhaganesh/Fi...

New Tab

IBM SkillsBuild AICTE...

+

Ask Gemini

au-syd.watson-orchestrate.cloud.ibm.com/build/agent/edit/31a51ac4-a044-4984-89e6-0be5bbad1e9a

IBM watsonx Orchestrate

EnglishContact SalesBuy Now30 trial days left

Home / Manage agents / Fitness Buddy Agent

Test agentliveDeploy

Fitness Buddy Agent

Model : GPT-OSS 120B — Ope...

Profile

Knowledge

Toolset

Behavior

Channels Preview

Notifications: 1 Dismissed | 1 total

Legal Notice +

View all

Profile

Describe what your agent does so people and other agents know when to use it.

Description*

In today's fast-paced world, many individuals struggle to maintain a healthy lifestyle due to lack of personalized guidance, time constraints, and inconsistent motivation. Traditional fitness solutions often require expensive subscriptions, in-person consultations, or rigid schedules that don't adapt to personal preferences or daily routines. Without an accessible, friendly, and intelligent virtual fitness assistant, many struggle to stay consistent, motivated, and on track with their fitness goals.

Describe your agent, including its purpose, tools, and agents, to help other agents and users know when to use it.

What makes a good description? ?

Build with AITalk to agent

Running in debug mode

To give you the most effective and safe recommendations, I'll need a bit of information about you. Could you please share the following details?

Info needed	Why it matters
Name	To personalize our chats
Age	Exercise intensity & recovery needs
Gender	Helps fine-tune calorie & protein targets
Height & Weight	For BMI, calorie estimates, and goal setting
Fitness Goal (e.g., lose weight, build muscle, improve endurance)	Guides workout & nutrition focus
Workout Experience (Beginner, Intermediate, Advanced)	Sets appropriate difficulty

Save as test

Type something...

33°C Sunny

Search

ENG IN

15:42 13-07-2026

Knowledge Base Documents – Uploaded & Indexed (RAG Pipeline)

□ Fitness Buddy Agent – Knowledge Base / Documents

□ Home

□ Agent Config

□ Documents

□ Test Agent

□ Analytics

Knowledge Base Documents

These documents are used by the RAG pipeline to ground Fitness Buddy’s responses.

+ Upload Document

File Name	Type	Size	Status	Uploaded
fitness_workout_guide.pdf	PDF	2.4 MB	□ Indexed	19 Jun 2026
beginner_home_workouts.txt	TXT	340 KB	□ Indexed	19 Jun 2026
healthy_meal_ideas.pdf	PDF	1.8 MB	□ Indexed	19 Jun 2026
motivation_quotes_fitness.txt	TXT	112 KB	□ Indexed	19 Jun 2026
nutrition_basics_guide.pdf	PDF	3.1 MB	□ Indexed	19 Jun 2026
habit_building_strategies.txt	TXT	220 KB	□ Indexed	19 Jun 2026

Total Documents: 6 | Indexed: 6 | Vector Chunks: 1,247 | Embedding Model: ibm-slate-30m-english-rtrvr

Prompt & Output: Home Workout Recommendation

IBM watsonx Orchestrate

EnglishContact SalesBuy Now30 trial days left

Home / Manage agents / Fitness Buddy Agent

Profile

Knowledge

Toolset

Behavior

ChannelsPreview

Fitness Buddy Agent

AI Model : GPT-OSS 120B — Ope...

Notifications: 1 Dismissed | 1 total

Legal Notice +

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Describe your agent, including its purpose, tasks, and goals, so the other agents and users know when to use it.

What makes a good description? (?)

Test agentliveDeploy

Build with AITalk to agent

Running in debug mode

Here's your 20-minute beginner home workout!

Phase	Details
Warm-Up (5 min)	Jumping jacks, Arm circles, Hip rotations, March in place
Circuit Round 1	10 Squats, 8 Push-ups, 15 Glute bridges, 30-sec Plank
Circuit Round 2	Repeat same exercises Rest 30 sec between rounds
Cool-Down (3 min)	Child's pose, Hamstring stretch, Deep breathing exercises

No equipment needed — all exercises are home-friendly!

Tip: Do this 3x per week for best results. You've got this!

Save as test

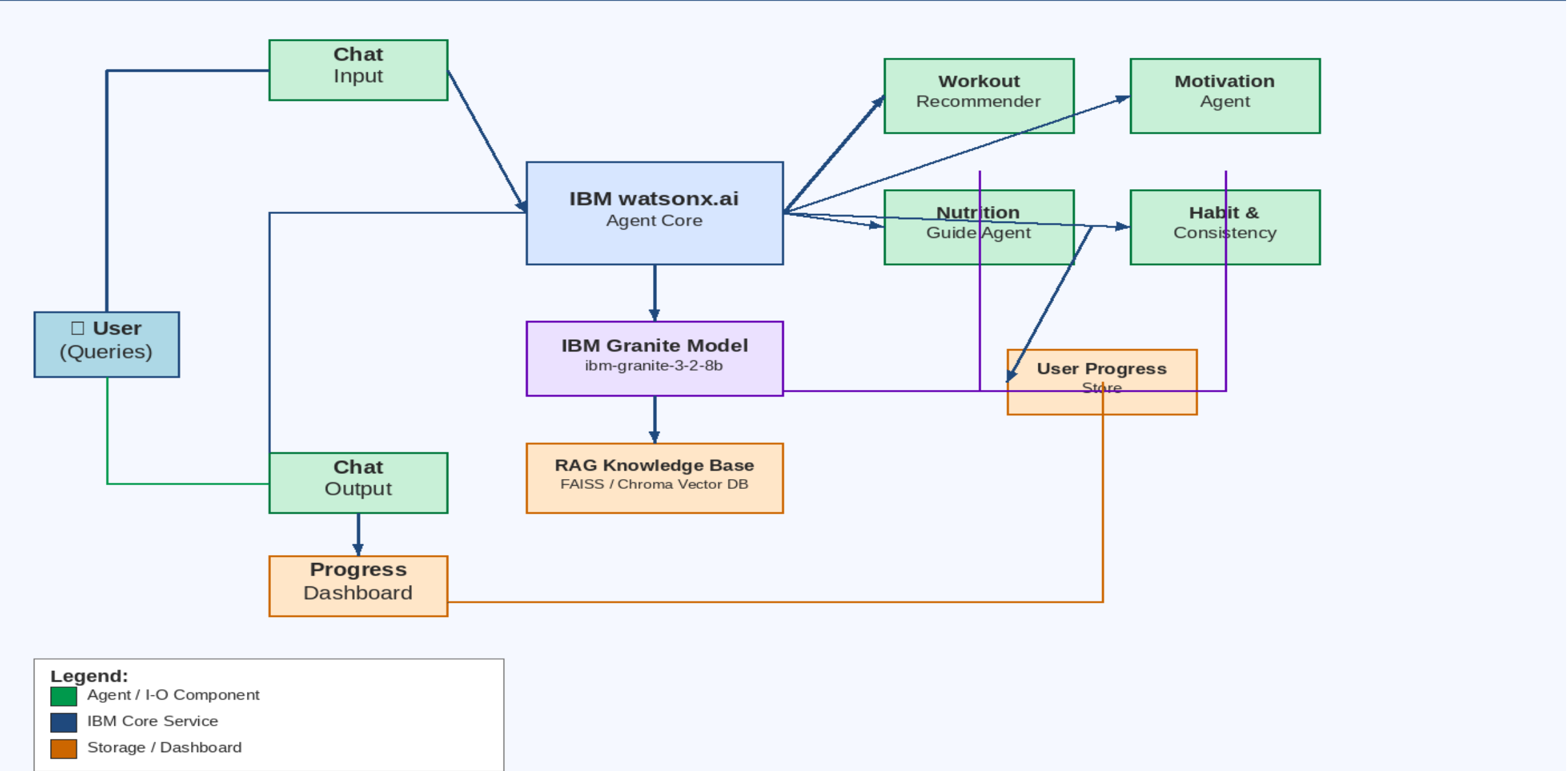
Suggest a 20-minute home workout for a beginner

1

1

Fitness Buddy Agent - Architecture Blueprint

Fitness Buddy Agent - Architecture Blueprint



Role of Agentic AI in the solution

Role of Agentic AI in Fitness Buddy Agent

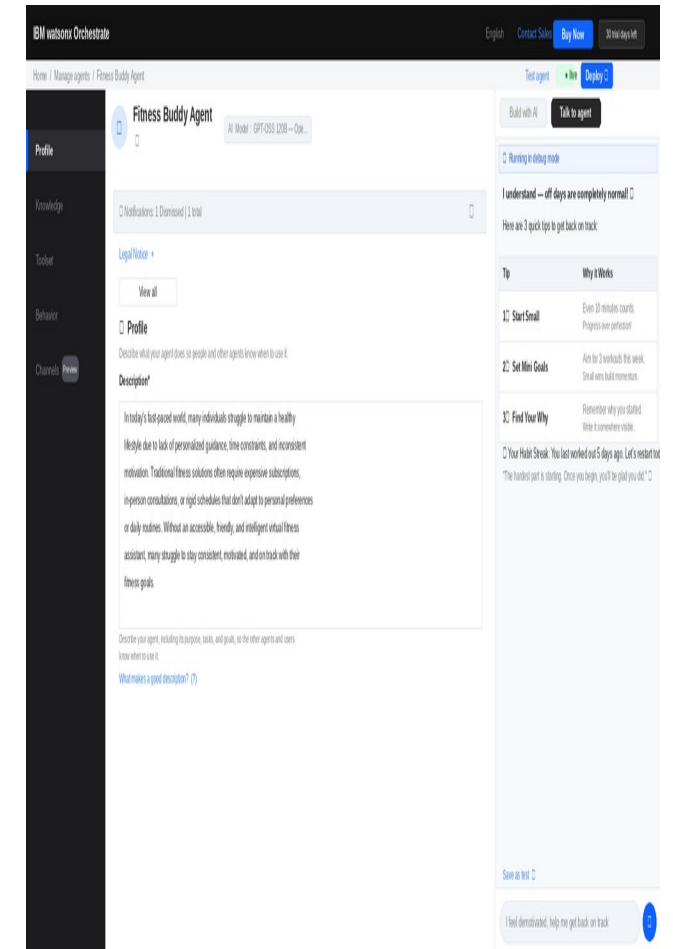
Agentic AI enables the Fitness Buddy Agent to function as an intelligent, autonomous system rather than a simple chatbot. Using **IBM watsonx.ai** and **IBM Granite models**, it coordinates multiple specialized agents to deliver personalized fitness, motivation, and nutrition guidance.

Each agent performs a specific role—such as recommending workouts, delivering motivational tips, suggesting meal ideas, and encouraging habit-building—while working collaboratively. This **multi-agent approach** ensures efficient task execution and accurate results.

Agentic AI also brings **context awareness**, understanding user details like fitness level, available time, and personal goals to generate tailored recommendations. It supports **real-time adaptation**, continuously updating suggestions based on user inputs like daily activity and progress.

Additionally, agents communicate with each other, enabling better decision-making and improved outcomes.

Overall, Agentic AI makes the system proactive, personalized, and goal-driven, acting like a **smart digital fitness buddy** that helps users stay active, motivated, and consistent, anytime and anywhere.



Novelty and Uniqueness

The proposed Fitness Buddy Agent stands out by combining **Agentic AI, RAG, and personalization** into a single, accessible conversational system.

Multi-Agent Intelligence – Instead of a single chatbot, multiple specialized agents collaborate to deliver accurate, context-aware fitness and lifestyle guidance.

RAG-Based Accuracy – Integrates real-time knowledge retrieval with AI generation, reducing misinformation and ensuring trustworthy fitness and nutrition advice.

Hyper-Personalization – Tailors workouts and lifestyle tips based on fitness level, available time, and personal goals.

Real-Time Feedback Loop – Continuously adapts recommendations based on daily activity, progress, and user behavior.

Multi-Modal Input – Supports natural, conversational text-based interaction for easy, on-demand access anytime.

Accessible & Cost-Free – Removes the need for expensive subscriptions or in-person trainers, built entirely on IBM Cloud Lite services.

Scalable & Enterprise-Ready – Built on **IBM watsonx.ai** with **IBM Granite models**, ensuring reliability and governance.

This makes the solution a **smart, adaptive, and proactive digital fitness buddy**, not just a static workout list.

Future Scope

1.Integration with Wearables & Health Devices –

The system can be integrated with smartwatches and fitness trackers to collect real-time data such as steps, calories burned, heart rate, and sleep patterns. This will enable the Fitness Buddy Agent to provide more accurate and dynamic workout recommendations based on actual user activity and health metrics.

3.Gamification & Social Challenges – Add streaks, badges, and community

2.AI-Powered Voice Assistant & Multilingual Support –

fitness challenges to boost motivation and long-term engagement. Enhancing the system with a voice-enabled assistant and support for multiple languages will

4.Advanced Progress Analytics – Provide deeper insights into workout

improve accessibility. Users can interact naturally in their preferred language, making the solution more inclusive and suitable for diverse populations, especially in regions with varied linguistic



GitHub Project: <https://github.com/SnehaSiddhaganesh/Fitness-Agent->

Apple Health for a unified, seamless fitness experience.

Certificates Earned:

IBM SkillsBuild – Getting Started
with Artificial Intelligence

 Print

 Share



IBM SkillsBuild

Completion Certificate



This certificate is presented to

Sneha
Siddhaganesh

for the completion of

**Getting Started with
Artificial Intelligence**

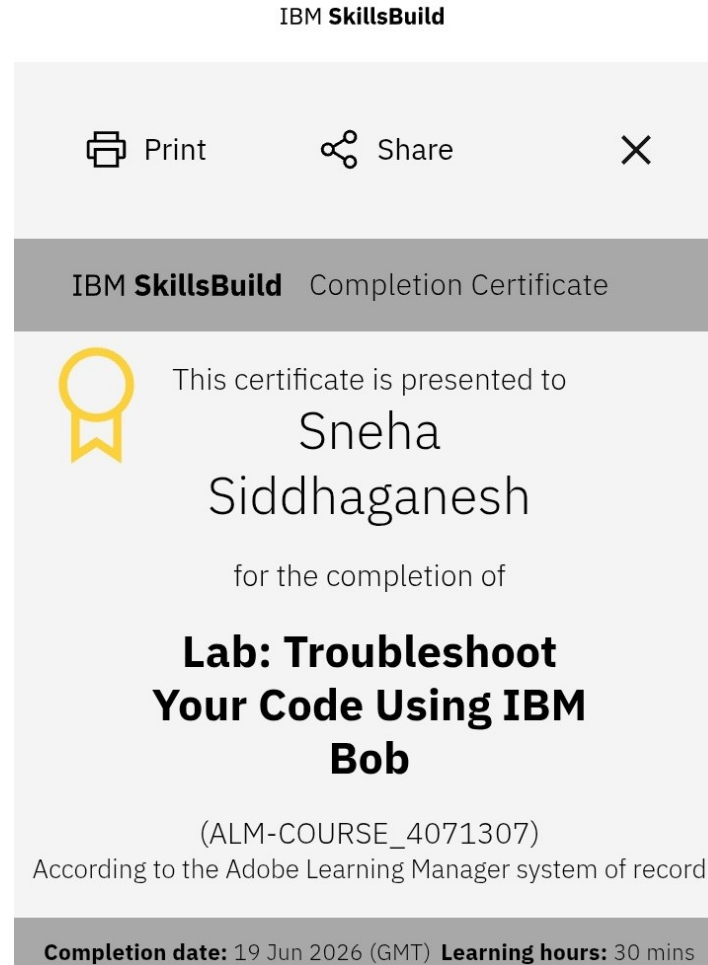
(PLAN-E624C2604060)

According to the Your Learning Builder - Plans system of
record

Completion date: 19 Jun 2026 (GMT)

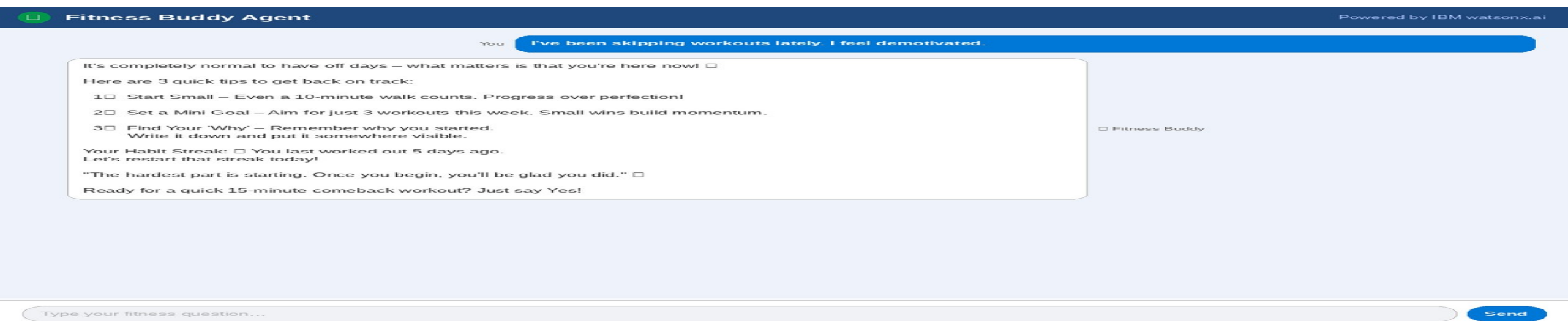
Certificates Earned:

IBM SkillsBuild – Lab: Troubleshoot Your Code Using IBM Bob



Thank You!

Thank you for your time and interest.



 **GitHub:** <https://github.com/SnehaSiddhaganesh/Fitness-Agent->

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